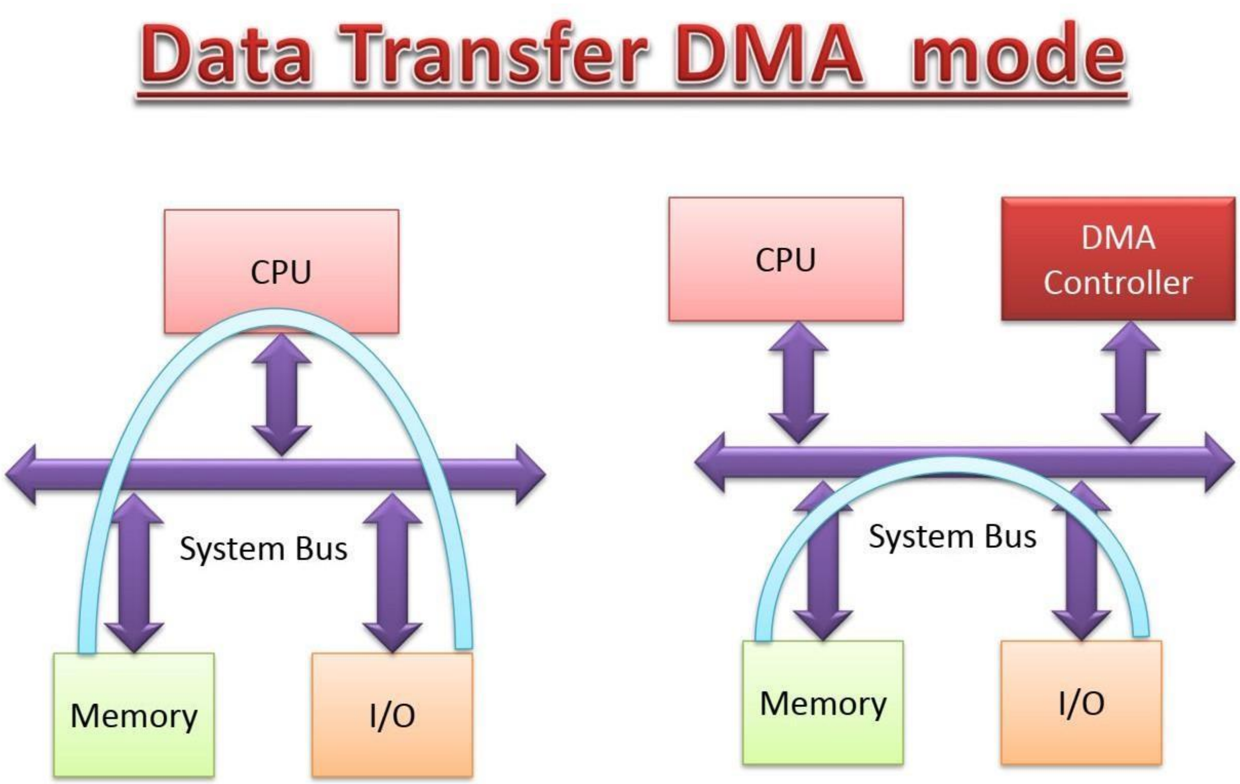
8237 (DMA Controller):

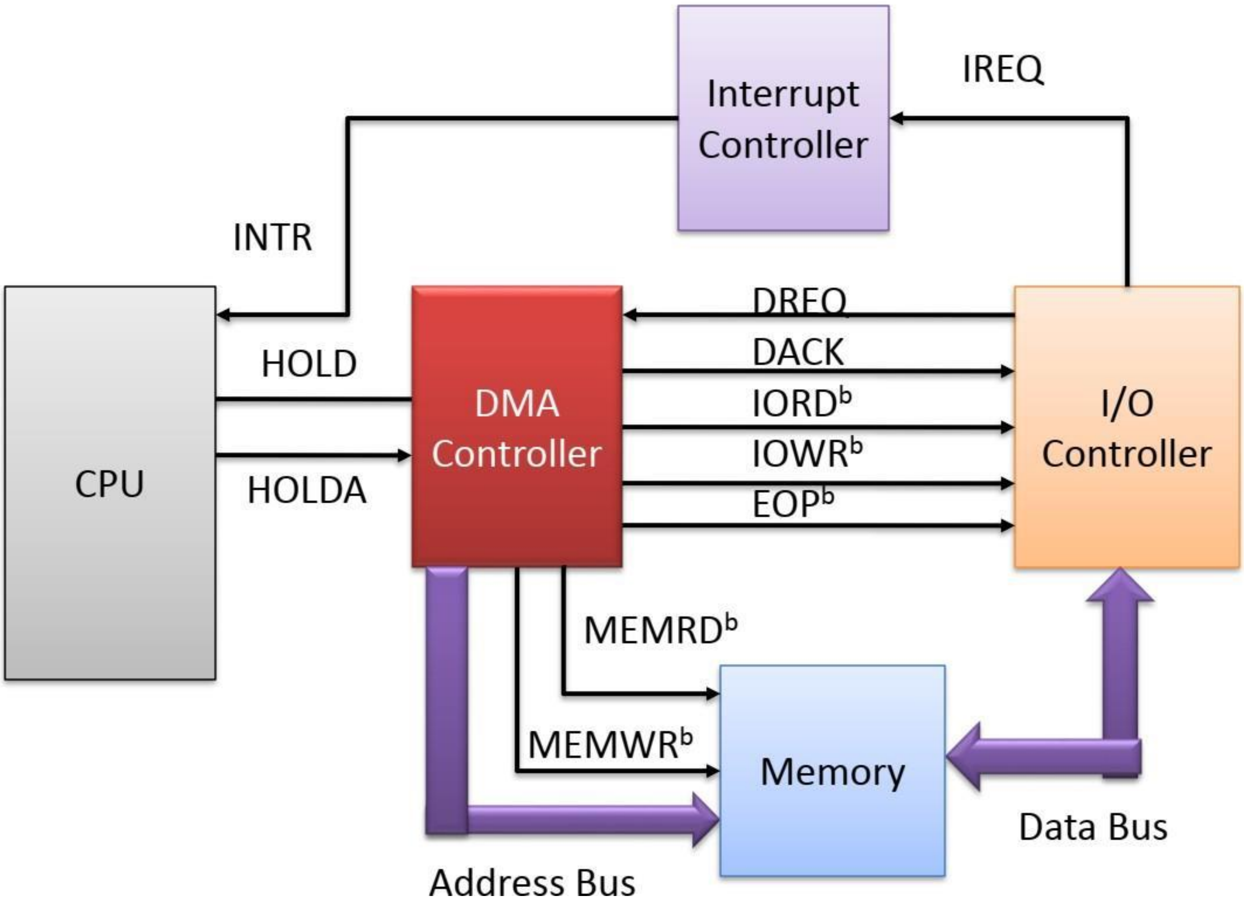
DMA: Direct memory access (DMA) is a feature of computer systems that allows certain hardware subsystems (such as I/O devices) to access main system memory independently without intervention from CPU.

It is a peripheral core for microprocessor systems. It controls data transfer between the main memory and the external systems with limited CPU intervention.

- It provides direct memory access where microprocessor is temporarily disable.
- We can transfer data very fast.



DMA Controller



DMA Controller interfaces with CPU, I/O Controller, Interrupt Controller and Memory.

HOLD = Hold signal

HOLDA = Acknowledge signal

Handshaking signals between DMA controller and CPU.

DREQ = DMA request signal,

DACK = DMA acknowledge signal,

IORDb = Input-Output read signal,

IOWRb = Input-Output write signal,

EOPb = End of operation signal

Handshaking signals between DMA controller and I/O controller.

MEMRDb = Memory read signal,

MEMWRb = Memory write signal

Handshaking signals between DMA controller and memory.